

Evolution of sectoral emissions and contributions to mortality from particulate matter exposure in the Asia-Pacific region between 2010 and 2015

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ABSTRACT

Anthropogenic emissions in the Asia-Pacific region have changed rapidly in recent years due to increasing industrialization and mobility, as well as the implementation of emission abatement controls. These changes are likely to have affected the region's burden of premature mortalities associated with exposure to fine particulate matter (PM_{2.5}). However, the contribution of each sector and effectiveness of different policy measures has not yet been quantified. As such, this study estimates changes in regional anthropogenic emissions by sector between 2010 and 2015. These changes are combined with an existing high-resolution emissions inventory to estimate emissions for the year 2015. Using a chemistry-transport model, we then estimate (i) the total contribution of each sector to premature mortality in 2015; and (ii) the effects of changes in each sector's contribution to total PM_{2.5}-driven premature mortalities between 2010 and 2015. We estimate that globally, 2,030,000 (95% CI: 1,770,000 to 2,280,000) PM_{2.5}-driven premature mortalities are attributable to anthropogenic emissions in 2015 in Asia-Pacific with the top three sources being the agricultural, industrial, and residential sectors. Between 2010 and 2015, excluding the effects of changes in population distribution or other social factors, sustained growth in economic activity led to an estimated 99,000 (95% CI: 81,000 to 130,000) additional premature mortalities annually, primarily across India, Indonesia, and Bangladesh. Simultaneously, changes such as electrification of railroads and newly-introduced abatement measures, e.g. China's Action Plan on the Prevention and Control of Air Pollution, region-wide adoption of Euro IV/V/VI-compliant road vehicles, and implementation of fuel quality standards, resulted in an estimated total reduction of 76,000 (95% CI: 65,000 to 100,000) annual premature mortalities, primarily across East Asia, including China and Japan. These opposing drivers add to a net change of an additional 22,000 (95% CI: 12,000 to 33,000) PM_{2.5}-driven annual premature mortalities between 2010 and 2015 associated with anthropogenic combustion emissions in Asia-Pacific.

1. Introduction

Anthropogenic emissions from activities such as combustion and agriculture directly and indirectly affect surface air quality, thereby negatively impacting human health. Past research has attributed ~6.4 million premature mortalities annually (Cohen et al., 2017) to human exposure to air pollution across the globe, exceeding the burden of other prominent global health threats such as malaria or tuberculosis (Apte et al., 2015; WHO, 2017). Of these mortalities, ~4.2 million are attributed to ambient particulate matter of a diameter of less than 2.5 µm (PM_{2.5}).

According to previous studies, over 50% of these premature

mortalities occur in the Asia-Pacific region, with two million premature mortalities annually being attributed to PM_{2.5} exposure in China and India in 2015 (Cohen et al., 2017). The significance of emissions in Asia-Pacific to human exposure to air pollution can be expected to increase for two reasons. First, emissions of PM_{2.5} and PM_{2.5} precursors in the region are estimated to have increased by 20–50% between 2000 and 2010 (Crippa et al., 2016) (Fig. 1). Second, the regional population grew significantly while becoming increasingly concentrated in polluted urban centers (UNPD, 2014). However, emissions controls measures, such as measures proposed in China's Action Plan on the Prevention and Control of Air Pollution (China State Council, 2013), counter these trends.

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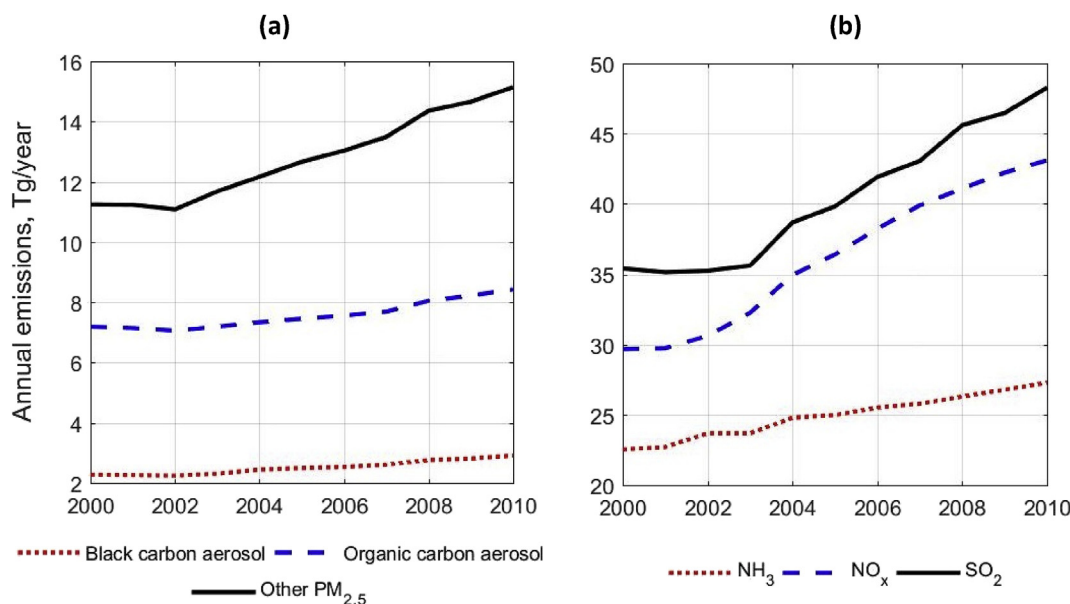


Fig. 1. Growth in emissions of (a) primary PM_{2.5} and (b) PM_{2.5} precursors in the Asia-Pacific region between 2000 and 2010 in Tg/year (NO_x as NO₂ mass basis). Data extracted from the Emission Database for Global Atmospheric Research (EDGAR) (Crippa et al., 2016).

This paper sets out to improve our understanding of the causes of and trends in Asia-Pacific air pollution. It is one of the first analyses to provide detailed insights into the air quality impacts of recent economic growth and emissions regulations across the Asia-Pacific region. For this purpose, it analyzes the sectoral drivers and effects of existing emissions controls as well as the spatial distribution and temporal evolution of regional and global impacts. The results not only enhance our understanding of the air quality issues in the region, but also help in evaluating the effectiveness of current mitigation strategies and in improving future strategies to counteract the regional and global burden of premature mortalities.

This paper addresses these objectives in two parts. In the first part, we develop a sectoral anthropogenic emissions inventory for the Asia-Pacific region that spans the 2010–2015 period, thereby fully covering the 12th Five-Year Plan of the People's Republic of China. Previous emissions inventories for the region were either produced prior to the period in question (e.g. Streets-2000 (Streets et al., 2003), INTEX-B (Zhang et al., 2009), MIX (Li et al., 2017), and EDGAR (Crippa et al., 2016)) or lack the sectoral, spatial, or temporal resolution (e.g. ECLIPSE (Amann et al., 2011; Klimont et al., 2017) and CEDS (Hoesly et al., 2018)) required for detailed analyses. Furthermore, although the MEIC regional emissions inventory (Zhang et al., 2009; Li et al., 2014; Zheng et al., 2014; Liu et al., 2015) has been continuously updated to the year 2015, these updates are only for Chinese emissions.

For this analysis, we use a composite approach to derive an inventory with fine sectoral resolution and emissions projections based on reported activity (UNSD, 2018) rather than projected activity indicators and emissions controls. We develop sector-specific inventory growth factors for the 2010–2015 period that incorporate sector-level activity indicators, published emissions standards, and information on emissions trends from the existing regional MIX 2010 and MEIC 2015 inventories (Zhang et al., 2009; Li et al., 2014, 2017; Zheng et al., 2014; Liu et al., 2015). To obtain emissions for the year 2015, we apply these growth factors to the EDGAR emissions inventory for 2010 (Crippa et al., 2016). The resulting inventory characterizes both the temporal evolution of regional anthropogenic emissions and the effect of existing emissions controls at a sectoral level.

In the second part of the paper, we use this inventory in the GEOS-Chem chemical transport model (CTM) to characterize the spatial distribution and temporal evolution in both regional and global air quality impacts by sector. We estimate the net change in regional and global

mortality due to anthropogenic emissions in Asia-Pacific between 2010 and 2015, the breakdown of changes in mortalities due to changes in emissions compared to change in demographics, and the effect of growth and emissions controls in eight sectors on mortality in each of 11 country groupings in the region, as well on mortalities exported to other global regions.

2. Methods

This section describes the methods used for (i) generating updated emissions in the Asia-Pacific region for the year 2015, (ii) modeling air quality and the resulting health impacts using the 2010 and 2015 emissions, and (iii) attributing the sectoral contributions to total global and regional mortalities in 2015 from emissions in the Asia-Pacific region.

2.1. Emissions

Emissions for 2015 are generated by applying a scaling method to the year-2010 baseline sectoral emissions obtained from the EDGAR 4.3.1 emissions inventory. The EDGAR emissions inventory, compiled by the European Commission Directorate General Joint Research Center (JRC) and the Netherlands Environmental Assessment Agency (PBL) (Crippa et al., 2016), is a comprehensive, gridded dataset of global anthropogenic emissions for 14 economic sectors (see Table 1) and 32 chemical species. It includes ammonia (NH₃), carbon monoxide (CO), nitrogen oxides (NO_x), 25 species of non-methane volatile organic carbon (NMVOCs), 3 species of primary PM_{2.5} including black and organic carbon, and sulfur dioxide (SO₂). It provides monthly-average, sector-specific emissions spanning a 40-year time series from 1970 to 2010, gridded spatially at a resolution of 0.1° × 0.1° degrees (longitude and latitude).

EDGAR v4.3.1 is computed using a technology-based emissions factor approach (Crippa et al., 2016). It has been widely applied in air quality modeling, including as the standard emissions inventory for CTMs such as GEOS-Chem, and has also been incorporated into the development of other emissions inventories, such as the HTAP, ECLIPSE, and CEDS global inventories and the REAS pre-2010 inventories for Asia (Janssens-Maenhout et al., 2015; Amann et al., 2011; Klimont et al., 2017; Hoesly et al., 2018; Kurokawa et al., 2013).

Out of 208 countries with sectoral emissions considered in EDGAR,

Table 1
Summary of sectors covered by the EDGAR 4.3.1 emissions inventory (Crippa et al., 2016). *CHP: Combined heat and power.

Sector	Description
Agriculture	Fertilizer use, animal husbandry, rice cultivation, waste burning
Aviation	Taxi, takeoff, climb, cruise, descent, and landing
Energy	Electrical power and heat generation (including cogeneration/CHP*)
Manufacturing	Manufacturing
Fossil fuel fires	Coal, oil, and gas fires
Industrial processes	Cement, lime, mineral, metal, food, and paper/pulp production
Fuel exploitation	Oil and gas extraction
Residential	Buildings/human habitation
Oil refineries	Oil refineries
Shipping	Domestic and international shipping
Waste processing	Emissions from waste incineration
Rail transportation	Emissions from rail vehicles and other ground transportation
Energy transformation	Fuel processing (fugitive emissions)
Road transportation	Emissions from road vehicles

25 from the Asia-Pacific region are selected for update to the year 2015. The countries under investigation span from East Asia and Southeast Asia, to (most of) the Indian subcontinent/South Asia (with the exception of Afghanistan and Pakistan). They are presented in Fig. 2. This region comprises an area with a population of 3.7 billion people in 2015.

For twelve of the EDGAR 4.3.1 sectors, we estimate the contribution of changes in emissions from each sector to changes in air pollution over the five-year study period. However, this level of detail is not presented for the remaining two sectors (aviation and fossil fuel fires). Due to their inherently transboundary nature, we treat aviation emissions as a special case. Instead of EDGAR data, we use the Federal

Aviation Administration (FAA) Aviation Environmental Design Tool (AEDT) inventories (FAA, 2016) and perform an analysis of the impact of aviation emissions relative to all other sectors in 2015 (see Section 3.3.1). Changes in impacts from changes in aviation emissions are not considered due to their relatively small contribution to net impacts and their global nature. Meanwhile the contribution of fossil fuel fires is not quantified due to the low total emissions (less than 1% of total emissions of any species considered) and low variance over time in this region.

While the 12 remaining sectors are treated distinctly for the purposes of emission modeling up to 2015, the manufacturing, industrial process, and waste processing sectors are combined into a single “industrial” sector when presenting results. Similarly, the fuel exploitation, oil refinery, and fuel transformation sectors are combined into a single “fuel processing” sector, giving eight separate sectoral groupings in total.

All chemical species constituting the EDGAR emissions inventory have direct or indirect impacts on aerosol chemistry, and are therefore considered in this study. This includes both primary PM_{2.5} (including BC and OC), and secondary PM_{2.5}, which is formed by precursor pollutants such as NH₃, NO_x, and SO₂. This study does not directly consider the effect of secondary organic aerosols (SOAs).

2.1.1. The scaling approach

In order to model changes in regional, sectoral emissions from 2010 to 2015 for the 25 countries under consideration, country-level information concerning variation in human activity with respect to each sector (e.g. fuel use for most combustion-driven sectors) as well as country-level changes in emissions standards and other abatement measures are combined into a scaling model. This model incorporates multi-dimensional underlying factors for emission trends, including multi-dimensional activity indicators (e.g. diesel fuel use by road transportation and oil refinery throughput) and abatement measures (e.g. emissions controls). The general form of this model is

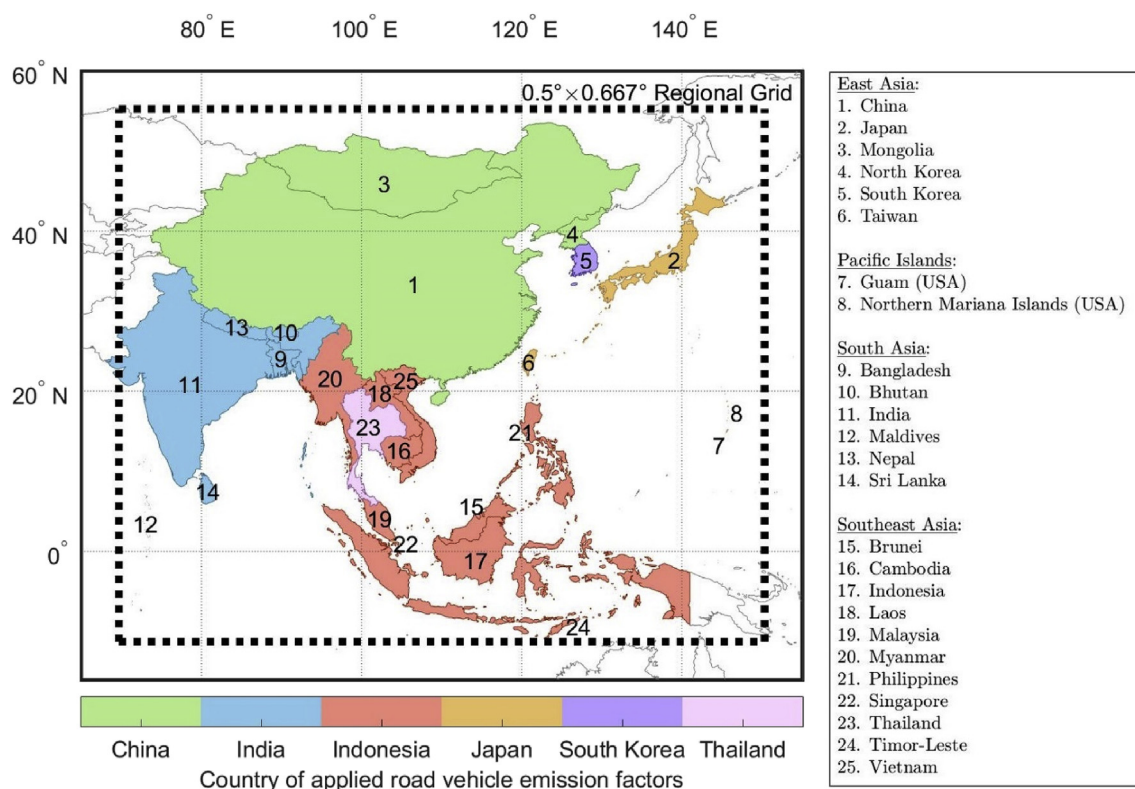


Fig. 2. Asia-Pacific countries considered in this study; countries are listed according to geographic sub-region and colored according to sub-region of applied road vehicle emissions factors (section 2.1.3).

$$E(c, n, s, m, y) = \left(\sum_{i=1}^{i_{\text{ind}}(c)} \alpha_i(c, n, s) r_i(c, n, s, m, y) \frac{V_i(c, n, y)}{V_i(c, n, 2010)} \right) E(c, n, s, m, 2010) \quad (1)$$

where:

E is the magnitude of emissions from a particular sector, country, species, and time
 c is the index for the emissions sector
 n is the index for the emitting country
 s is the index for the emitted species
 m is the index for the month of the year (1–12)
 y is the index for the year (2010–2015)
 i_{ind} is the number of activity indicators pertinent to each sector
 α is an apportioning factor weighting each activity indicator of a sector (Section 2.1.3)
 r is an abatement factor for each activity indicator of a sector (Section 2.1.4)
 V_i is an activity indicator for a sector (Section 2.1.2)
 The inputs required to parametrize this model are discussed in the following sub-sections.

2.1.2. Sector-specific activity indicators

Activity indicators for each EDGAR sector are compiled from multiple sources. For most combustion-driven sectors, fuel use indicators are taken from the United Nations Statistics Division Energy Statistics Database (UNSD-ESD) (UNSD, 2018). This database details each country's energy flows and balances on an annual basis. It incorporates usage statistics of over 50 different industrial substances and fuels by more than 40 different activities, capturing all combustion-driven sectors included in EDGAR. In addition, the database provides direct activity statistics for some sectors, such as refinery throughput for oil refineries, as well as oil and gas production for the fuel exploitation and energy transformation sectors.

For sectors, which are not combustion-driven or for which activities are not well-characterized in the UNSD-ESD database (e.g. agriculture, industrial process emissions, and waste processing), alternative activity indicators are used. All activity indicators for the agricultural sector are taken from statistics and indices provided by the United Nations Food and Agricultural Organization (FAO) (FAO, 2018). Activity indicators for industrial process emissions are derived from three sources, namely the U.S. Geological Survey (USGS) for production statistics of aggregates, metals, and minerals (USGS, 2018); the BASF data on regional growth statistics of chemical production (BASF, 2017); and FAO statistics for food processing and paper production (FAO, 2018). Waste processing emissions are scaled with urban population data obtained from the UN Population Division via the World Bank Databank (UNPD,

2014).

Table 2 summarizes specific activity indicators and their sources for both combustion-driven and other sectors.

2.1.3. Apportioning factors

Of the sectors listed in Table 1, only three (oil refineries, waste processing, and rail transportation) are modeled using a single activity indicator. For the remaining nine sectors, we consider changes in emissions to be driven by multiple activity indicators. For such sectors, apportioning factors are computed. These factors determine the fraction of 2010 emissions attributable to each activity indicator in order to scale emissions using equation (1). For the agriculture and industrial process sectors, a breakdown of total sectoral emissions by sub-activity is provided by EDGAR, from which apportioning factors for each emission species can be computed directly. For the remaining sectors, apportioning factors must be computed on the basis of sector-specific emissions factors and 2010 activity indicators. The factors by activity indicator are obtained from multiple sources as detailed in Table 3. The factors for all activity indicators are then aggregated into a single factor by emissions sector as shown in equation (2).

$$\alpha_i(c, n, s) = \frac{f_i(c, n, s, m, 2010) V_i(c, n, 2010)}{\sum_{i=1}^{i_{\text{ind}}(c)} f_i(c, n, s, m, 2010) V_i(c, n, 2010)} \quad (2)$$

where f_i is the sector-specific emission-factor for each activity indicator.

While for most sectors, the emissions factors described in Table 3 apply region-wide, we use country-specific emissions factors for road transportation in 6 major countries in this region: China, India, Japan, Indonesia, South Korea, and Thailand. These emissions factors are applied sub-regionally as denoted by the coloring of countries in Fig. 2.

2.1.4. Abatement factors

As shown in equation (1), the computation of changes in emissions between 2010 and 2015 considers abatement factors, which model the impacts of new emissions regulations and controls implemented between 2010 and 2015. These abatement factors are calculated as

$$r_i(c, n, s, m, y) = \frac{f_i(c, n, s, m, y)}{f_i(c, n, s, m, 2010)} \quad (3)$$

Abatement factors are used for the energy, manufacturing, and residential sectors in China and for the shipping and road transportation sectors region-wide. We consider these sectors and regions in an effort to model emissions reductions associated with (i) China's Action Plan on the Prevention and Control of Air Pollution, in effect since 2013 (Zheng et al., 2018; China State Council, 2013); (ii) regulations by the International Maritime Organization mandating a reduction of fuel sulfur content in international marine bunker fuel from 4.5% to 3.5% from 2012 onwards (Soares and Santos, 2015); and (iii) country-specific

Table 2

List of activity indicators by sector.*CHP: Combined heat and power.

Sector	Activity Indicator(s)	Source(s)
Agriculture	FAO Food Production Index, FAO Livestock Production Index, fertilizer use, rice production	FAO
Energy	Anthracite, bituminous coal, fuel oil, lignite, natural gas use by power/heat/CHP* plants	UNSD-ESD
Manufacturing	Anthracite, bagasse, bituminous coal, blast furnace gas, diesel, fuel oil, fuelwood, kerosene, LPG, natural gas use by manufacturing facilities	UNSD-ESD
Industrial processes	Cement, lime, steel production	USGS
	Chemical production	BASF
	Paper production, FAO Food Production Index	FAO
Fuel exploitation	Crude oil, natural gas production	UNSD-ESD
Residential	Anthracite, bituminous coal, diesel, fuel oil, fuelwood, Kerosene, LPG, natural gas, vegetal waste use by commercial facilities/households	UNSD-ESD
Oil refineries	Refinery throughput	UNSD-ESD
Shipping	Diesel, fuel oil use by domestic navigation	UNSD-ESD
	Diesel, fuel oil use in intl. Marine bunkers	UNSD-ESD
Waste processing	Urban population	World Bank
Rail transport	Diesel use by railroads	UNSD-ESD
Energy transform	Crude oil, natural gas production	UNSD-ESD
Road transport	Diesel, gasoline use by road vehicles	UNSD-ESD

Table 3

Sources of sector-specific emissions factors.*CHP: Combined heat and power.

Sector	Emissions factors for:	Source(s)
Energy	Anthracite, bituminous coal, lignite combustion in power/heat plants Fuel oil, natural gas combustion in power/heat/CHP* plants	Zhao et al. (2010) and Xu et al. (2017) U.S. EPA AP-42 (U.S. EPA, 1995) U.S. EPA NEI-2014 (U.S. EPA, 2016)
Manufacturing	Anthracite, bagasse, bituminous coal, blast furnace gas, diesel, fuel oil, fuelwood, kerosene, LPG, natural gas combustion by manufacturing facilities	Lewis (1997)
Fuel Exploitation	Crude oil, natural gas extraction/processing/transportation	U.S. EPA NEI-2014 (U.S. EPA, 2016)
Residential	Anthracite, bituminous coal, diesel, fuel oil, fuelwood, kerosene, LPG, natural gas, vegetal waste combustion by commercial facilities/households	Winnes and Fridell (2009)
Shipping	Diesel, fuel oil combustion in marine engines	Lewis (1997)
Energy Transform	Crude oil, natural gas extraction/processing/transportation	Country-specific emissions standards
Road Transport	Diesel, gasoline combustion in automotive engines (country-specific)	

emissions regulations implemented in the road transportation sector ([International Council on Clean Transportation and DieselNet, 2018](#)).

In the case of the Chinese energy sector, we apply abatement measures for NO_x, primary PM_{2.5}, and SO₂ emissions. NO_x emissions are estimated according to [Liu et al. \(2016\)](#), who find a 60% reduction in the emissions factor of NO_x from Chinese coal-fired plants between 2011 and 2015 as a result of the widespread implementation of selective catalytic reduction technologies. This reduction is represented as a linear ramp in the NO_x emissions factor for Chinese coal-fired power plants over the period under investigation. Abatement measures are also applied for primary PM_{2.5} and SO₂ emissions from the Chinese energy, manufacturing, and residential sectors. For these emissions, abatement factors are derived to match trends in the MEIC regional inventory ([Zhang et al., 2009](#); [Li et al., 2014](#); [Zheng et al., 2014](#); [Liu et al., 2015](#)), as documented by [Zheng et al. \(2018\)](#).

For the regional shipping sector, abatement factors are derived for SO₂ emissions from international shipping. This is done in accordance with International Maritime Organization regulations mandating a reduction of fuel sulfur content in international marine bunker fuel from 4.5% to 3.5% from 2012 onwards ([Soares and Santos, 2015](#)).

For road transportation, the implementation of emissions controls is modeled on a detailed level. Abatement factors are derived for all emissions species, and for all six major countries in this region. The road vehicle emissions factors are then applied sub-regionally. Trends in emissions factors include uptake time and are estimated from nationally published vehicle emissions and fuel sulfur standards for both diesel and gasoline ([International Council on Clean Transportation and DieselNet, 2018](#)).

Although the region-wide shipping and road transportation sectors, and the Chinese energy, manufacturing, and residential sectors are the only ones where emissions controls are explicitly modeled through application of abatement factors, controls that affect overall fuel use are captured in the sector-specific activity indicators. This includes transitions between different types of fuels (e.g. from coal to natural gas in the case of energy emissions in Myanmar) and electrification (e.g. in the case of the railroad sector in China).

2.1.5. Other considerations

While the speciation of NMVOCs for the updated inventory is provided by EDGAR, no such speciation was provided for PM_{2.5}, other than of black and organic carbon (BC and OC). As a result, the remaining component of PM_{2.5} was segmented into individual species including primary ammonium, nitrates, and sulfates as well as trace metals on the basis of a speciation profile developed by [Fu et al. \(2013\)](#) for 24 different combustion sources that cover the 12 EDGAR sectors considered in the bulk of this study. Furthermore, the conversion of OC tracer into total organic matter is computed using OM/OC factors derived for GEOS-Chem simulations by [Philip et al. \(2014\)](#).

Furthermore, although the scaling approach described was applied consistently across all 12 sectors, it was found to be ill suited for modeling changes in emissions from the energy sectors of Cambodia

and Sri Lanka as a result of major changes in both fuel consumption patterns and spatial patterns of emissions. These changes result from the start of operations of the first large-scale coal-fired power plants in these countries (Sihanoukville I and II plants in Cambodia, and Lakvijaya plant in Sri Lanka). For these cases, we include an additional emissions source at the location of each plant, using data for coal-fired plants of similar scale in the GPED 2010 Global Power Emissions Database to define their emissions profiles ([Tong et al., 2018](#)).

2.2. Air quality modeling

The GEOS-Chem chemical transport model (CTM) is used to simulate the effect of changes in sectoral emissions from 2010 to 2015 on surface-level concentrations of PM_{2.5}. GEOS-Chem uses nested Eulerian grids to simulate atmospheric processes including emissions, chemistry, transport, and deposition. It provides hourly estimates of the resulting ground-level concentration of PM_{2.5}. First developed by [Bey et al. \(2001\)](#), the version of the GEOS-Chem CTM applied here uses the Kinetic PreProcessor chemical solver ([Damian et al., 2002](#)) for gas-phase and heterogeneous chemistry. The MARS-A model ([Binkowski and Roselle, 2003](#)) is applied for calculation of aerosol thermodynamic equilibrium and partitioning of key species between the gas- and aerosol-phase. For this study, the forward modeling component of GEOS-Chem adjoint version 35f ([Henze et al., 2007](#)) is used. The model is modified to accept global 2010 EDGAR 4.3.1 emissions and the Asia-Pacific emissions updated up to 2015.

The GEOS-Chem CTM is used to simulate two domains. The first is a global domain with 4° latitude × 5° longitude resolution, from which we obtain global impacts of modeled sectoral emissions and emissions changes in the Asia-Pacific region. The second is a nested 0.5° × 0.667° grid centered on the Asia-Pacific region (see [Fig. 2](#)), providing a high-resolution estimate of regional impacts. The finer regional grid extends from 10.5° S to 55.5° N and 69.5° E – 150.5° E, with boundary conditions specified by the global runs. Although this regional grid excludes a small part of the western Indian state of Gujarat, the population of the excluded portion is less than 0.1% of the total population of India. It therefore has a small effect on the results of this study, which are presented at a national level. Across all simulations, a 47-layer hybrid sigma-pressure grid is used, with a model top of 1 Pa (80 km altitude). This version of GEOS-Chem is driven by meteorological data from the GEOS data assimilation system, provided by the NASA Global Modeling and Assimilation Office.

Outside of the Asia-Pacific region, the unmodified 2010 EDGAR emissions inventory is applied in global simulations for both year-2010 and 2015 runs. Where available, this is superseded by regional emissions inventories such as the U.S. EPA NEI-2011 inventory for the United States of America ([U.S. EPA, 2015](#)). Although this approach excludes modeling the change in background PM_{2.5} in the Asia-Pacific region between 2010 and 2015, this is not expected to significantly affect results, which are primarily driven by local contributors and as the background only accounts for ~20% of total exposure, with the

effect of changes in this on changes to mortalities caused by local emissions over the 5 year period expected to be even lower.

Wildfire emissions for each year are provided by the GFED4 wildfire inventory distributed by Oak Ridge National Laboratory (Randerson et al., 2017). Non-anthropogenic emissions are calculated during the simulation based on meteorological data. Lightning NO_x emissions are calculated based on convective cloud depth and corrected by observations (Murray et al., 2012). Emissions of mineral dust are calculated using the Dust Entrainment and Deposition (DEAD) mobilization scheme developed by Zender et al. (2003). Finally, sea-salt emissions are calculated using the model developed by Alexander (2005).

To analyze the contribution of each sector to the change in global and regional PM_{2.5} exposures between 2010 and 2015, a series of 18-month GEOS-Chem simulations is conducted. The first 6 months of each run are discarded as a model spin-up period to remove the influence of initial conditions and the remaining 12 months are averaged to provide the annual-average concentration of PM_{2.5}. To quantify the total global and regional population exposure to PM_{2.5} attributable to sectoral emissions from the Asia-Pacific region in 2015, as well as the change over the 2010–2015 period, both global and regional domain runs are first conducted with all Asia-Pacific sectoral emissions set to 2015 values. These emissions are then eliminated or reverted to 2010 levels respectively for each of the 12 sectors considered. As such, in addition to the year 2015 baseline run, two global domain runs and two regional domain runs are conducted for each sector. In the first pair of global/regional runs, emissions for the sector analyzed are completely eliminated; in the second pair of runs, emissions are reverted to 2010 emissions.

Computing the difference between annual-average ground-level PM_{2.5} distributions from runs with all emissions set to 2015 values and runs with individual sectors set to 2010 gives each sector's contribution to the change in global/regional PM_{2.5}-driven mortalities between 2010 and 2015. Applying the same method to the runs with emissions eliminated by sector allows computation of each sector's contribution to total PM_{2.5}-driven mortalities attributable to year-2015 emissions in the Asia-Pacific region.

2.3. Health impact assessment

To compute premature mortalities attributable to each economic sector from the ground-level PM_{2.5} distributions obtained from GEOS-Chem, we apply the Integrated Exposure Response (IER) function developed by Burnett et al. (2014). The IER is used to estimate the effect of increased PM_{2.5} exposure on mortality rates from chronic obstructive pulmonary disease (COPD), ischemic heart disease (IHD), lower respiratory infections (LRI), lung cancer, and cerebrovascular disease (strokes). Parameters for each IER function are adopted from the 2016 WHO Global Burden of Disease (GBD) (Gakidou et al., 2017). The mean and 95% confidence intervals of IERs for each of the five health endpoints are presented in Fig. A3. A single curve is used to characterize relative risk of COPD, LRI, and lung cancer at all ages. Two sets of 15 age-specific curves (at five-year intervals between the ages of 25 and 95, with decreasing relative risk at increasing age) are considered to characterize relative risk of IHD and strokes. These curves provide an estimated relative risk (RR) value for a specific group for a given exposure to PM_{2.5}. Comparing relative risk at different exposure levels and combining it with age- and location-specific baseline mortality data (see below) allows estimation of the likely increase in incidence of a specific disease.

Relative risks obtained by applying the IERs to ground level PM_{2.5} concentrations from the GEOS-Chem results are used to determine the fraction of total incidence for each health endpoint per grid-cell attributable to emissions from each economic sector. This is done by first computing the PM_{2.5}-attributable fraction of total incidence for all sectors applying the GEOS-Chem runs with all sectors eliminated or set to 2010 levels, and then apportioning the result by sector on the basis of

change in PM_{2.5} attributable to each sector from the GEOS-Chem runs with individual sectors eliminated or set to 2010 levels. This method ensures that the non-linearity of the Burnett IERs, which yield lower marginal relative risk with increasing PM_{2.5} exposure, does not result in differential treatment of exposure resulting from emissions from any individual sector. This two-step process is summarized in equations (4a/4b) and (5).

2.4. The total anthropogenic contribution for all sectors in 2015 is given by

$$\Delta I_h(x, y) = I_h \left(1 - \frac{RR_{h, \text{off}}(x, y)}{RR_{h, 2015}(x, y)} \right) \quad (4a)$$

The change in anthropogenic contribution due to all sectors between 2010 and 2015 is given by

$$\Delta I_h(x, y) = I_h \left(1 - \frac{RR_{h, 2010}(x, y)}{RR_{h, 2015}(x, y)} \right) \quad (4b)$$

where:

I_h is the total incidence of health endpoint h

ΔI_h is the PM_{2.5}-attributable fraction of I_h due to all sectors

$RR_{h, \text{off}}$ is the relative risk with all Asia-Pacific sectoral emissions eliminated

$RR_{h, 2010}$ is the relative risk with all Asia-Pacific sectoral emissions at 2010 values

$RR_{h, 2015}$ is the relative risk with all Asia-Pacific sectoral emissions at 2015 values

x, y are (2-D) spatial indices of the ground-level GEOS-Chem PM_{2.5} distribution

ΔI_h is then apportioned by sector as shown in equation (5)

$$\Delta I_{h,c}(x, y) = \Delta I_h(x, y) \frac{\Delta PM_{2.5,c}(x, y)}{\sum_{i=1}^{i_{\text{sect}}} \Delta PM_{2.5,i}(x, y)} \quad (5)$$

where:

$\Delta I_{h,c}$ is the incidence of health endpoint h attributable to each sector c

i_{sect} is the number of sectors

$\Delta PM_{2.5}$ is the change in PM_{2.5} exposure by eliminating or reverting emissions from individual sectors

For this study, country-specific incidence rates (i.e. baseline mortality rates) of the five identified health endpoints (I_h) for 2010 and 2015 at all age brackets relevant to the WHO GBD 2016 IERs are obtained from the WHO GBD 2015 study (Wang et al., 2016). These incidences are applied in Equations (4a/4b) to compute fractional incidences attributable to each sector, which are subsequently multiplied by the relevant population (ages 25 + for IHD and strokes, and all ages for COPD, LRI, and lung cancer) in each grid-cell to compute mortalities. The population distribution is considered at 30 arc-second resolution globally as provided in the LandScan 2010, 2015 datasets (ORNL, 2010; ORNL, 2016).

Premature mortalities are quantified both regionally and globally. Regional premature mortalities are computed for the 10 most populous countries in the Asia-Pacific region (Bangladesh, China, India, Indonesia, Japan, Myanmar, Philippines, South Korea, Thailand, and Vietnam) and for the remainder of the region combined. Global impacts are quantified both in total and for 8 separate regions. Four regions are defined outside Eurasia according to continental grouping (Africa, North America, South America, and Oceania). Meanwhile Eurasia is split into four regions (Asia-Pacific, the Commonwealth of Independent States, Europe, and the Middle East) in order to distinguish near-field and far-field impacts. A map of these global regions is presented in Fig. A2.

Table 4

Evolution of emissions of primary PM_{2.5} and PM_{2.5} precursors between 2010 and 2015 in (a) China, (b) India, (c) Indonesia, and (d) other Asia-Pacific countries in Tg (species)/year (NO_x as NO₂ mass basis).

(a) China					(b) India				
	Primary PM _{2.5}	NH ₃	NO _x	SO ₂		Primary PM _{2.5}	NH ₃	NO _x	SO ₂
2010	13	14	24	30	2010	6.6	6.8	8.2	9.4
2011	13	14	26	31	2011	7.0	7.0	8.8	10
2012	13	15	25	28	2012	7.2	6.6	9.0	11
2013	12	15	24	25	2013	7.5	6.5	9.6	12
2014	12	15	22	22	2014	7.7	6.8	9.9	13
2015	11	15	21	20	2015	8.0	7.0	11	14

(c) Indonesia					(d) Other Asia-Pacific				
	Primary PM _{2.5}	NH ₃	NO _x	SO ₂		Primary PM _{2.5}	NH ₃	NO _x	SO ₂
2010	2.1	1.9	2.2	2.0	2010	4.7	4.6	8.5	6.6
2011	2.2	2.0	2.6	2.7	2011	4.8	4.8	8.6	6.6
2012	2.3	2.1	2.8	3.0	2012	4.9	4.9	8.5	6.6
2013	2.3	2.1	2.7	3.0	2013	4.9	5.2	8.4	7.1
2014	2.4	2.1	2.5	3.1	2014	5.0	5.4	8.3	7.1
2015	2.5	2.3	2.8	3.3	2015	5.0	5.4	8.3	7.3

3. Results

3.1. Emissions trends between 2010 and 2015

As in [Dedoussi and Barrett \(2014\)](#), changes in ground-level distribution and population exposure to PM_{2.5} are principally driven by changes in emissions of total primary PM_{2.5} (consisting of BC, OC, primary sulfates, nitrates, ammonium, and trace metals) and the three major chemical precursors of secondary PM_{2.5} (NH₃, NO_x, and SO₂). Estimated annual emissions of each of these chemical species between 2010 and 2015 are shown in [Table 4](#) for the three highest-emitting countries China, India, and Indonesia (representing 80–83% of total anthropogenic NH₃, NO_x, primary PM_{2.5}, and SO₂ emissions in the region), as well as for the remainder of the region combined.

The emissions trends vary by species and location. These trends are attributable to both changes in human activity and fuel use by sector, as well as government-driven measures to introduce emissions controls through both legislative and technological means. This is also visible in [Fig. 3](#), as the changes in NH₃ emissions from the agricultural sector and NO_x emissions from three emitting sectors (energy, railroads, and road transportation) have unique spatial patterns.

The broad increases in ammonia emissions visible in [Fig. 3\(a\)](#) are driven by growth in the agricultural sector, specifically by changes in fertilizer use and livestock production. Growth is highest in South and Southeast Asia, especially with the major agricultural producers Bangladesh, Indonesia, and Vietnam (34%, 22%, and 31% growth respectively). Conversely, agricultural ammonia emissions from Japan and Malaysia are found to fall by 9% and 7% respectively between 2010 and 2015. These reductions are driven by changes in fertilizer use, as reported by the UN FAO ([FAO, 2018](#)). Changes in NH₃ emissions from India exhibit a mixed pattern, with a 9% decrease in annual emissions modeled between 2011 and 2013 caused by a dip in fertilizer use. This pattern may be attributable to changes in Indian government policies on agricultural subsidies, including an 11% cut in subsidies for nitrogen-based fertilizers between 2012 and 2013 ([Nayak and Jadhav, 2012](#)), which may have affected fertilizer usage in the short-term.

Changes in emissions of NO_x, primary PM_{2.5}, and SO₂ vary across the Asia-Pacific region ([Fig. 3\(b\)](#)). NO_x emissions for India and Indonesia increase by 29% and 27% respectively, primarily driven by increased activity in the energy, manufacturing, and residential sectors. This represents a total increase in annual NO_x emissions of 3.0 Tg, more than 36% of total NO_x emissions in 2015 from all other Asia-Pacific

countries. Annual Chinese NO_x emissions instead fall, decreasing by 3.7 Tg between 2010 and 2015 – a 15% decrease relative to 2010 levels. This is likely caused by the introduction of selective catalytic reduction (SCR) technologies into Chinese coal-fired power plants, resulting in a 60% reduction of corresponding emissions factors according to [Liu et al. \(2016\)](#). Despite an increase of 17% in the use of coal by the Chinese energy sector over the five years under investigation, the introduction of SCR caused a net reduction of 40% of NO_x emissions from this sector, and results in emissions of NO_x from China falling continuously after reaching a maximum in 2011. This trend in Chinese NO_x emissions is consistent with existing literature by [Liu et al. \(2017\)](#), which used satellite observations to identify a peak in column-total NO_x concentrations over China in 2011.

NO_x emissions from the rest of the Asia-Pacific region are found to decrease by 2–3% between 2010 and 2015, which relates to a 0.2 Tg reduction in overall annual emissions. This decreasing trend is especially noticeable in South Korea, where annual NO_x emissions from road vehicles fell by 70%, or 0.5 Tg, between 2010 and 2015. This decrease was due to the implementation of more stringent vehicle emission standards. This policy-driven effect on the road transportation sector is visible across most of the region with the exception of Malaysia, Indonesia, and the Philippines ([Fig. 3\(d\)](#)).

Changes in emissions of both total primary PM_{2.5} and SO₂ follow a similar polarized pattern to that observed for energy NO_x in [Fig. 3\(b\)](#), with reductions in China and a net increase in the remainder of the region. Reductions of these species in China are driven by the energy, manufacturing, and residential sectors, and attributable to several abatement measures adopted as part of China's Action Plan on the Prevention and Control of Air Pollution in effect since 2013 ([Zheng et al., 2018](#); [China State Council, 2013](#)). We estimate that emissions of PM_{2.5} and SO₂ from the Chinese energy sector fell by 25% and 50%, respectively, over this period due to a range of measures including increasing use of flue gas desulfurization in power plants, and promoting the use of higher-quality coal with lower ash and sulfur content ([Zheng et al., 2018](#)). In the case of the Chinese manufacturing sector, other measures such as the phase out of small, coal-fired industrial boilers in urban areas, desulfurization of emissions from large industrial boilers through sorbent injection, installation of electrostatic precipitators to remove primary PM_{2.5} emissions from manufacturing plants, and replacement of old manufacturing plants with new equipment of higher combustion efficiency, result in reductions of PM_{2.5} and SO₂ emissions by 28% and 41% respectively ([Zheng et al., 2018](#)). Reductions of 23% in primary PM_{2.5} emissions and 19% in SO₂ emissions from the Chinese residential sector are driven by the introduction of improved technology such as cleaner-burning stoves ([Zheng et al., 2018](#)). Furthermore, shifts in fuel use towards cleaner-burning forms of coal (e.g. briquettes) also aid emissions reduction in both of these sectors in China ([Zheng et al., 2018](#)). Combined, abatement measures applied in China to these three sectors result in reductions of 17% (2.2 Tg) and 35% (10.5 Tg) in primary PM_{2.5} and SO₂ emissions, respectively, over the period under investigation. These reductions exceed growth in emissions of these two species in the rest of the region, despite high net growth (21% and 47% respectively) in India.

Apart from these major sectoral drivers of emissions changes, many other drivers contribute to the total change in emissions between 2010 and 2015. For example, the effect of more stringent road vehicle emissions standards is not limited to reduction of just NO_x emissions, but also of emissions of CO, NMVOCs, and primary PM_{2.5}. More stringent fuel quality standards implemented for the road transportation sector similarly reduce SO₂ emissions attributable to road transportation. Furthermore, changes in diesel fuel use by the rail transportation sector lead to different trends in emissions of all species by this sector across the region, as shown in [Fig. 3\(c\)](#). While strong growth in diesel use by railways leads to growth in South Asian rail emissions, specifically by 15% in India and 70% in Bangladesh, declining diesel use in other parts of the region such as East and Southeast Asia, attributed to

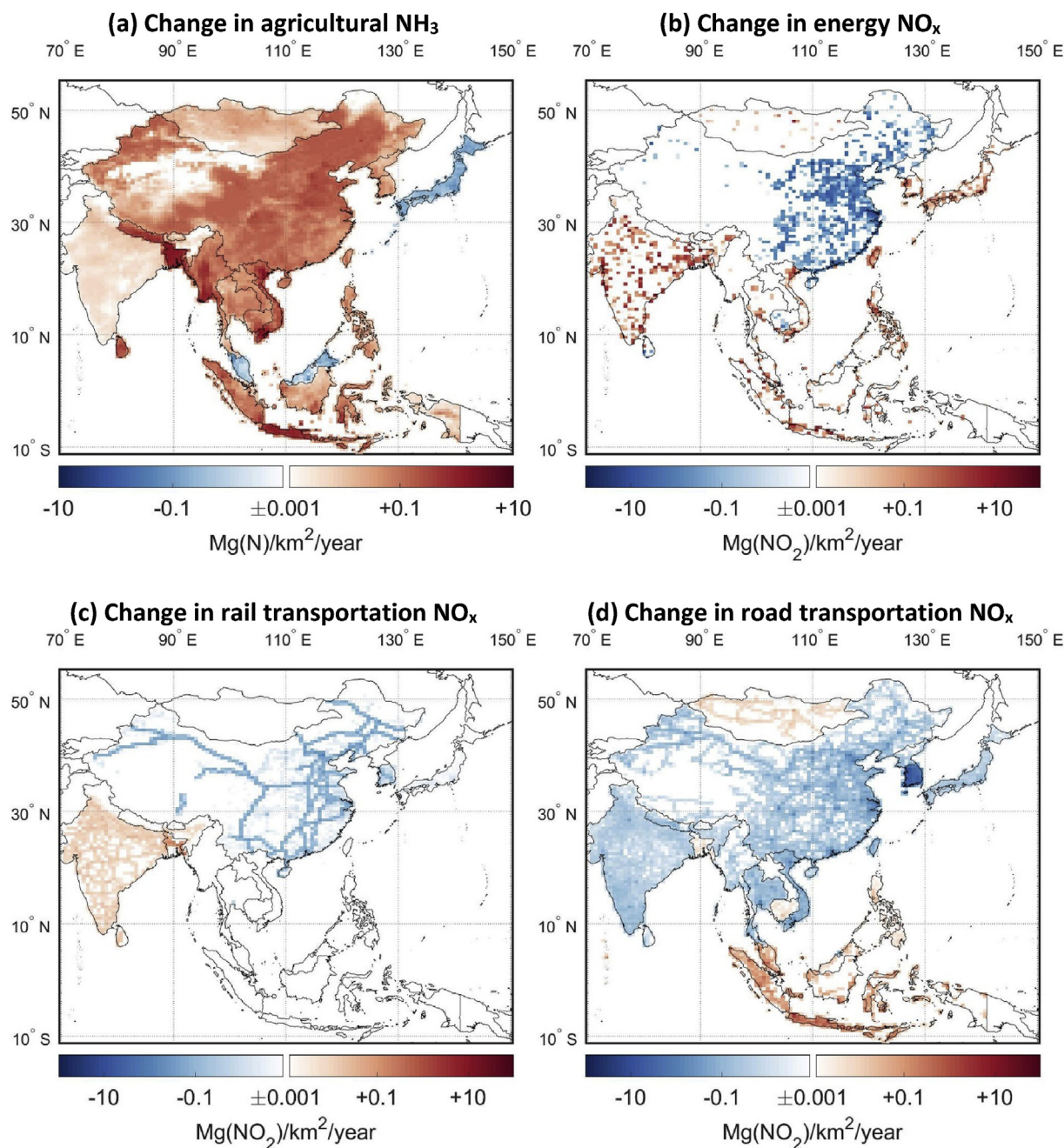


Fig. 3. Spatial distributions of changes in (a) agricultural NH_3 emissions in units of $\text{Mg(N)}/\text{km}^2/\text{year}$, and changes in NO_x emissions in units of $\text{Mg}(\text{NO}_2)/\text{km}^2/\text{year}$ from the (b) energy, (c) railroad, and (d) road transportation sectors in the Asia-Pacific region between 2010 and 2015.

electrification of railways and phase-out of operating diesel-fired rolling stock, leads to emission reductions from the rail sector of 41%, 7.0%, and 39% in China, Japan, and South Korea respectively.

3.2. Validation

We compare simulated ground-level $\text{PM}_{2.5}$ from GEOS-Chem to satellite-derived distributions from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) instruments (NASA Earth Observations, 2018). For these comparisons, regional wildfire emissions were updated to 2015 values to yield a better match against the observed spatial distribution of $\text{PM}_{2.5}$. This is not the case for the sectoral impact evaluation runs used to compute premature mortalities, as this study does not examine changes in $\text{PM}_{2.5}$ -driven mortalities due to non-anthropogenic sources such as wildfires. These comparisons also exclude the

contributions of dust and sea-salt to ground-level $\text{PM}_{2.5}$, as these are not measured in the MODIS results (Van Donkelaar et al., 2018). Fig. 4 shows the simulated and satellite-derived annual average ground-level $\text{PM}_{2.5}$ across the region in both 2010 and 2015, as well as the change in $\text{PM}_{2.5}$ between 2010 and 2015.

GEOS-Chem successfully captures the broad spatial pattern and magnitude of ground-level $\text{PM}_{2.5}$ over the Asia-Pacific region in both 2010 and 2015, including high values over the Chinese industrial heartland in eastern China and the Indo-Gangetic plain in northern India. Several individual features, such as the distinct hotspot of $\text{PM}_{2.5}$ over the Chongqing-Chengdu metropolitan areas in South-Central China are also captured in the GEOS-Chem simulations. Comparison of the change in ground-level $\text{PM}_{2.5}$ between the two years shows that the GEOS-Chem simulations capture the broad trends: increases in India; decreases across large parts of China; and increases over the majority of

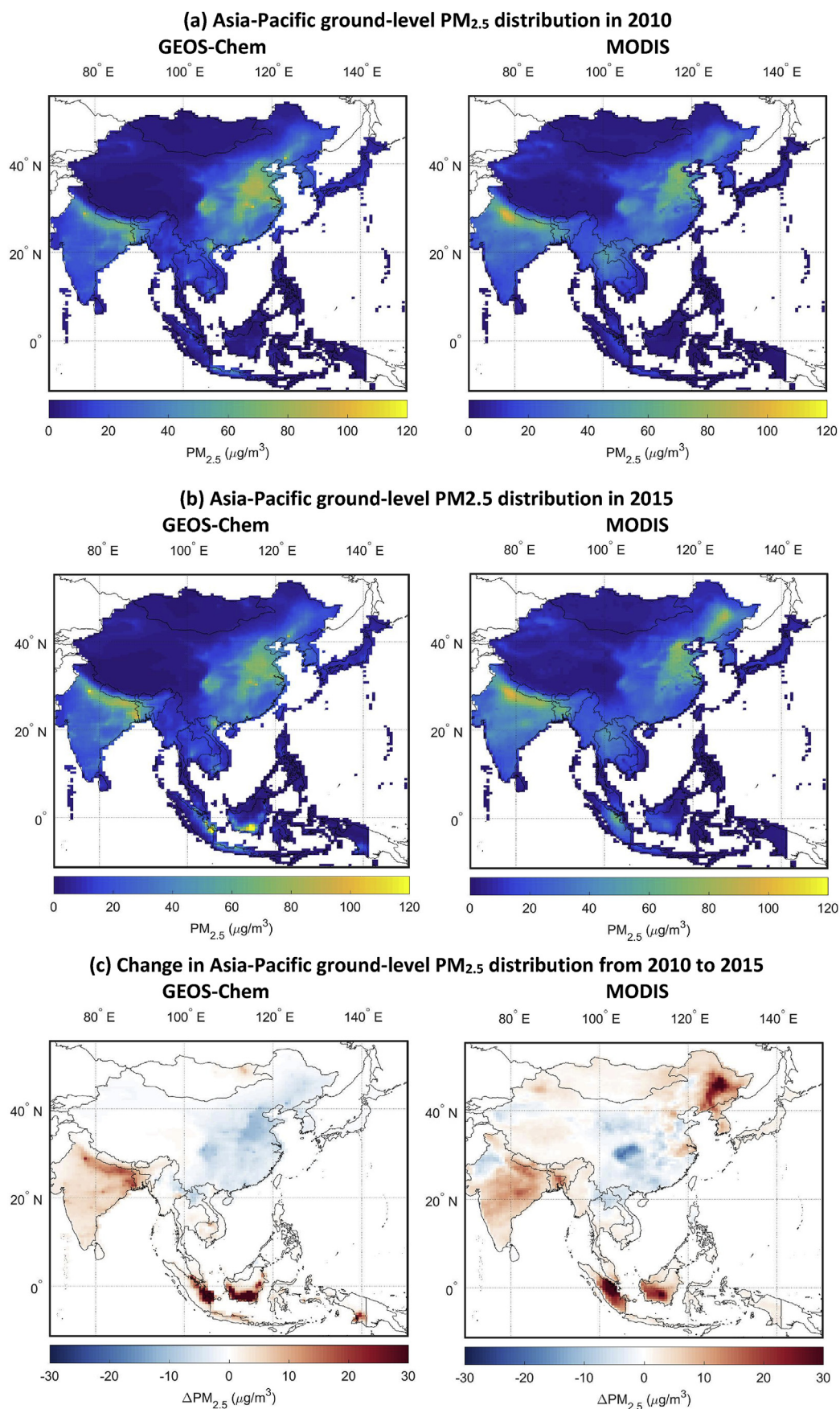


Fig. 4. Comparison of ground-level PM_{2.5} from NASA MODIS satellite observations and GEOS-Chem simulations for **(a)** 2010, **(b)** 2015, and **(c)** the change from 2010 to 2015.

the Southeast Asia sub-region.

While no major differences are observed between the spatial patterns of ground-level PM_{2.5} according to GEOS-Chem and MODIS in 2010, discrepancies between ground-level PM_{2.5} in 2015 according to GEOS-Chem and MODIS results are particularly prominent in parts of Southeast Asia including Indonesia, Malaysia, Brunei, and the Philippines. The disagreement in these parts of the Asia-Pacific are likely driven by non-anthropogenic sources that lie outside the scope of this study. In particular, results are thought to be affected by the limited spatial accuracy of the GFED4 wildfire inventory used to model wildfires in 2015 across Southeast Asia.

Although these disagreements between GEOS-Chem and MODIS can cause high localized differences in PM_{2.5}, our focus is on the change in annual-average population exposure to PM_{2.5}. We find that for 2010, the error in GEOS-Chem values of population-weighted PM_{2.5} relative to MODIS values is +4.0 µg/m³ (+11%), well within typical error bounds for modeling studies in this region (Hu et al., 2017; Geng et al., 2017). For the five most populous countries in this region which account for more than 85% of the regional population, the error is: +11.6 µg/m³ in China; -4.3 µg/m³ in India; +11.3 µg/m³ in Indonesia; -4.7 µg/m³ in Bangladesh; and +3.6 µg/m³ in Japan. The remainder of the region has a mean error of +3.2 µg/m³ between modeled and satellite-derived results.

In the case of population-weighted ground-level PM_{2.5} in 2015, the region-wide error between the GEOS-Chem and MODIS results decreases by 36% from 2010 to +2.6 µg/m³ (+7.1%), primarily driven by reductions in the country-specific errors for China and India, at +6.1 µg/m³ and -2.1 µg/m³ respectively. Across the remaining populous countries, the error between GEOS-Chem and MODIS increases in both Bangladesh and Indonesia, at -8.2 µg/m³ and +15.0 µg/m³ respectively, but decreases compared to 2010 in both Japan as well as the remainder of the region, with 2015 errors of +1.5 µg/m³ and 2.2 µg/m³ respectively. Table A4 lists the population-weighted PM_{2.5} both region-wide and from each of the listed countries as obtained from GEOS-Chem and MODIS for both 2010 and 2015.

Overall, these country-level errors for both 2010 and 2015 suggest that the GEOS-Chem results for this study over-predict the population exposure to PM_{2.5} in East and Southeast Asia, and under-predict in South Asia relative to MODIS results. However, both the region-wide and country-specific discrepancies for both years remain less than 15 µg/m³, or less than 7.5% of the total range of the Burnett IERs used to compute premature mortalities. For the ambient PM_{2.5} concentrations prevalent in populated areas of this region, this error translates to an error of 10% or less in relative risks calculated from the IERs, as shown in Fig. A3. This level of error is lower than the 95% confidence intervals computed for the mortality results of this study.

The comparison of change in ground-level PM_{2.5} between 2010 and 2015 as per GEOS-Chem and MODIS in Fig. 4(c) also exhibits several differences, particularly across Southeast Asia, where GEOS-Chem overestimates PM_{2.5} over areas due to limitations in modeling year-2015 wildfires, and also along the Asia Pacific-Russia border stretching from Mongolia to northern China. We attribute the latter to two potential causes. Firstly, changes in non-regional sources that lie outside the scope of this study, including changes in both anthropogenic emissions and wildfire activity in Russia may affect this border region. Secondly, the methodology used to update Asia-Pacific emissions from 2010 to 2015 in this study assumes uniform changes in terms of activity indicators and abatement measures within any given country. However, past studies (Koukouli et al., 2016) indicate greater effectiveness in adoption of abatement technologies by the energy sector in the south of China compared to the north. As a result, the effect of abatement measures on energy sector emissions in northern China between 2010 and 2015 may have been over-predicted, and this may also explain the observed discrepancy in 2010–2015 change in ground-level PM_{2.5} distributions in this area according to GEOS-Chem and MODIS. Table A5 lists the change in population-weighted PM_{2.5} both region-wide and

from each of these countries as obtained from GEOS-Chem and MODIS.

3.3. Health impacts

The changes in emissions from the Asia-Pacific region between 2010 and 2015 lead to changes in human exposure to ambient PM_{2.5} and therefore in rates of premature mortality. Each sector's contribution to the change in both global and regional mortalities between 2010 and 2015 is presented in this section. The sectoral contribution to total mortalities is quantified to put the effect of the changes between 2010 and 2015 in context, and to identify shifts in the relative impact of individual sectors. Furthermore, the effect of abatement measures on changes in the mortality burden are assessed and sectoral targets for future mitigation strategies are identified.

Globally, 1.78 million (95% CI: 1.56 million to 2 million) and 2.03 million (95% CI: 1.77 million to 2.28 million) PM_{2.5}-driven premature mortalities are attributed annually to anthropogenic emissions from the Asia-Pacific region in 2010 and 2015 respectively. In both years 98% of these premature mortalities are found to occur in the Asia-Pacific region itself, and the remaining 2% are exported to other regions. However, out of the total increase in annual global premature mortalities associated with Asia-Pacific emissions (250,000 mortalities (95% CI: 210,000 to 280,000)), only 9%, or 22,000 mortalities (95% CI: 12,000 to 33,000), are attributable to changes in ambient concentrations of PM_{2.5}, with the remaining 91% being attributable to changes in demographics, including population growth, migration, aging, and change in total incidence of the health endpoints linked to PM_{2.5} exposure. Fig. 5 summarizes the breakdown of total mortalities due to each of these drivers. As drivers other than PM_{2.5} concentrations are outside the scope of regulation by emissions controls, we only report the change in mortalities due to change in PM_{2.5} exposure, and consistently use 2015 population distribution and demographics unless otherwise stated. Changes in mortality between 2010 and 2015 discussed from this point onwards therefore refer to the change in expected mortality under the assumption of population distributions and demographics which are fixed to 2015 data.

3.3.1. Regional impacts

Of the 1,990,000 annual PM_{2.5}-driven mortalities resulting from Asia-Pacific anthropogenic emissions and occurring within the Asia-Pacific region in 2015, the majority occur in China and India, with 910,000 (95% CI: 790,000 to 1,030,000) and 710,000 (95% CI: 620,000 to 810,000) mortalities respectively. Together, these two countries, which contain around 70% of the regional population, account for more than 80% of the regional burden of PM_{2.5}-driven mortalities in 2015. These results are consistent with previous calculations of PM_{2.5}-driven mortalities for these countries (Lelieveld et al. (2015); Cohen et al. (2017)) (Table A1). A further 270,000 (95% CI: 230,000 to 300,000) mortalities per year occur in Bangladesh, Indonesia, Japan, Thailand, and Vietnam collectively. The remaining regional burden is dominated by other relatively populous countries such as Myanmar, the Philippines, and South Korea.

3.3.1.1. Impacts by sector. Fig. 6(a) and Table A6 show the total PM_{2.5}-driven mortalities per year in each country by sector. Emissions from the agricultural sector, including agricultural waste burning, are the leading source of PM_{2.5}-driven premature mortalities both at a regional and national level for most of the receptor countries with the exception of Vietnam, where residential emissions have a roughly equal effect. Ambient PM_{2.5} from agricultural emissions cause 590,000 (95% CI: 510,000 to 660,000) mortalities per year across the region, more than 30% of the total regional burden. This is followed by the industrial, residential, and energy sectors, which have similar net impacts totaling 1,230,000 (95% CI: 1,070,000 to 1,380,000) across all three sectors.

The total impact of global aviation (including non-Asian cruise emissions) is found to be small relative to other sectors, with 4600 (95%

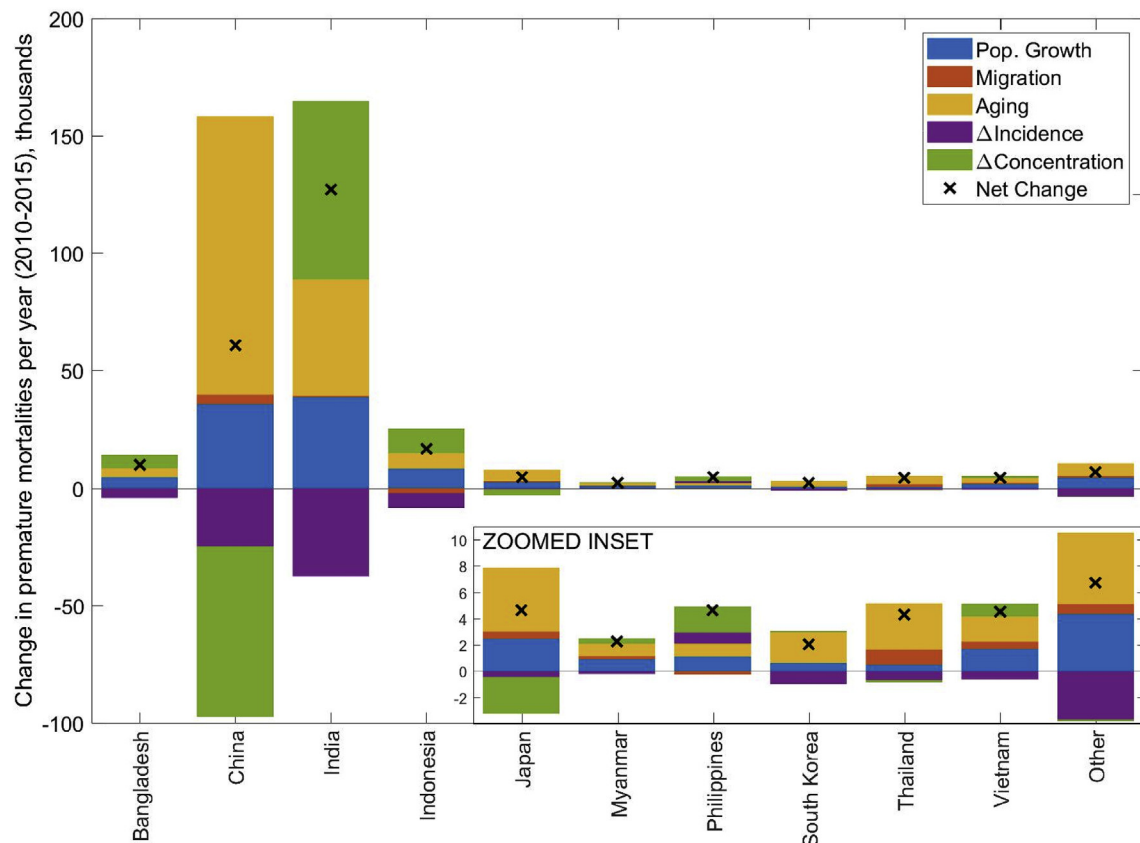


Fig. 5. Total change in PM_{2.5}-driven premature mortalities between 2010 and 2015 due to all drivers in Asia-Pacific receptor countries.

CI: 3900 to 5900) annual premature mortalities being attributable to this sector. The impact of global aviation emissions on premature mortalities associated with degraded air quality in the Asia-Pacific region is therefore less than 0.3% of the total impact from regional emissions, and less than half of the total impact from regional railroad emissions in 2015. Due to the relatively small impact, the global nature of aviation, and the indirect connection between local policy change and global changes in aviation emissions, we do not further quantify impacts of aviation on Asia-Pacific air quality or changes in this sector over the period analyzed.

Regional growth in PM_{2.5}-attributable mortalities between 2010 and 2015 is driven by opposing changes in emissions from individual countries, whose drivers differ from those of total regional impacts in 2015 (Fig. 6(b) and Table A7). Across South and Southeast Asia, sustained economic activity growth leads to 96,000 (95% CI: 78,000 to 130,000) additional annual premature mortalities in 2015 compared to 2010, primarily impacting India, Indonesia, and Bangladesh. However, other sectors and parts of the region undergo emissions reductions due to falling activity rates from structural changes such as electrification of railroads and emissions controls from regulations. Such regulations include China's Action Plan on the Prevention and Control of Air Pollution (China State Council, 2013), which targeted the Chinese energy, industrial, and residential sectors, as well as region-wide adoption of Euro IV/V/VI-compliant road vehicle emission and fuel quality standards. These abatement measures cause a total reduction of 76,000 (95% CI: 64,000 to 100,000) annual premature mortalities, primarily across East Asia, including China, Japan, and South Korea. These opposing trends lead to a net increase in regional PM_{2.5}-driven mortalities of 20,000 (95% CI: 14,000 to 23,000), or 1.4%, due to changes in ambient PM_{2.5} concentrations between 2010 and 2015.

3.3.1.2. Impacts by country. Breaking down these impacts by country rather than sector, the greatest increase in premature mortality impacts

between 2010 and 2015 is found in India, increasing by 76,000 (95% CI: 61,000 to 100,000) annual mortalities (13% relative to 2010). This increase stems primarily from the energy and industrial sectors, two of the four leading contributors to baseline Indian PM_{2.5} in 2015. Increased emissions from these sectors result in an additional 26,000 (95% CI: 21,000 to 35,000) and 39,000 (95% CI: 32,000 to 53,000) annual mortalities respectively, or over 85% of the net increase in PM_{2.5}-attributable mortalities in India. Although agricultural and residential emissions are similarly large contributors to baseline Indian PM_{2.5}, they show no significant growth compared to industrial and energy emissions.

Impacts in China follow the opposite trend. Our results suggest that, between 2010 and 2015, premature mortalities in China decrease by 73,000 (95% CI: 62,000 to 98,000), or 8.6% relative to 2010. The reduction in PM_{2.5}-driven premature mortalities in China observed between 2010 and 2015 is likely due to the emissions controls imposed on the energy, industrial, and residential sectors in this five-year period. In the Chinese energy sector, the introduction of SCR, transition to cleaner sources of coal with lower ash and sulfur content, and expansion of flue gas desulfurization in Chinese power plants (Zheng et al., 2018) reduced NO_x, primary PM_{2.5}, and SO₂ emissions, resulting in 30,000 (95% CI: 25,000 to 40,000) fewer mortalities in 2015 than 2010 in China alone. As electricity generation grew by 38% between 2010 and 2015, from 4200 TWh to 5800 TWh (National Bureau of Statistics of China, 2017), mortalities per unit of electricity generated fell by 50%, from 24 mortalities/TWh in 2010 to 12 mortalities/TWh in 2015 (considering only the direct air quality impacts of power plant emissions).

For the Chinese industrial sector, the phase-out of small coal-fired industrial boilers and outdated equipment, as well as application of desulfurization and particulate filtering in other industrial facilities, contributes to a reduction of 25,000 (95% CI: 18,000 to 37,000) annual mortalities in 2015 compared to 2010. Lastly, introduction of technologies such as clean-burning stoves as well as cleaner forms of fuels in

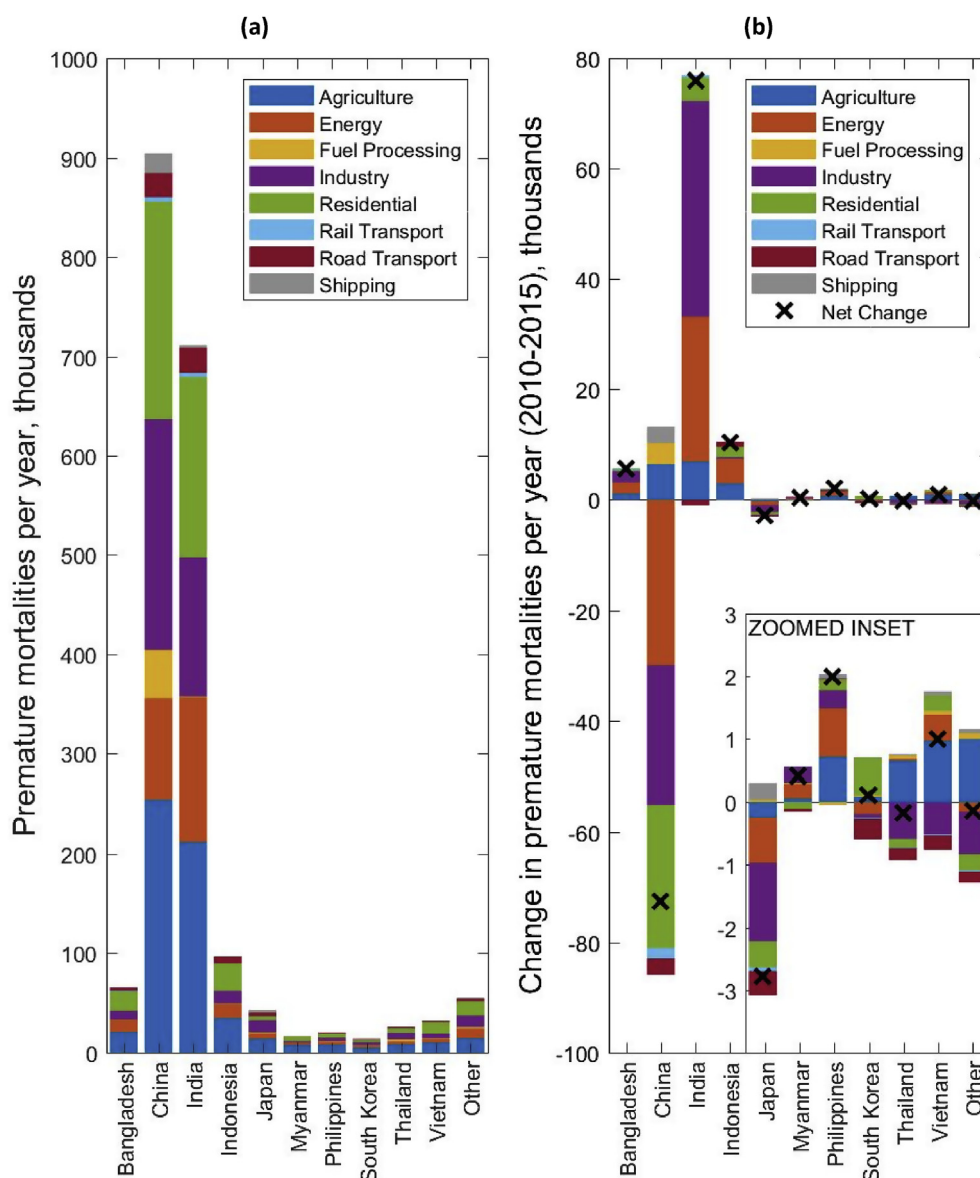


Fig. 6. Sectoral contributions to (a) total annual PM_{2.5}-driven mortalities in Asia-Pacific countries in 2015 and (b) change in annual PM_{2.5}-driven mortalities in Asia-Pacific countries between 2010 and 2015 in mortalities/year (based on 2015 population distributions and demographics).

the Chinese residential sector contributes to 26,000 (95% CI: 22,000 to 35,000) fewer annual mortalities in 2015 than 2010. The reductions in mortalities achieved in any of these three sectors offset the growth in mortalities in China from all other sectors. Furthermore, the reduction in mortalities from the Chinese energy and residential sectors also exceeded the rise in mortalities from the corresponding sector in India. In the case of the industrial sector, growth in mortalities in India exceeds reductions in China by 14,000 annual mortalities over this five-year period.

The imposition of emission controls for these sectors in China has also had transboundary benefits. We estimate reductions in premature mortalities attributable to these sectors in neighboring countries such as South Korea (10% fewer premature mortalities per year due to energy emissions relative to 2010) despite 6–8% growth in South Korean energy sector emissions between 2010 and 2015.

3.3.1.3. Transportation emissions. Regional impacts from ground-based transportation have fallen between 2010 and 2015. Reductions in emissions from rail transportation result in 1300 (95% CI: 1100 to 1800) fewer premature mortalities in 2015 compared to 2010 across

the region. The trend in annual premature mortalities due to rail transportation by country mirrors the trend observed for the energy sector, with reductions observed in China, and increases observed in India. Although no abatement measures are explicitly modeled for the rail transportation sector, the reduction of sector-attributable premature mortalities in China by 1800 (95% CI: 1600 to 2500) premature mortalities in 2015 than in 2010 could be driven by increasing electrification and modernization of the Chinese railway locomotive fleet. The region-wide reduction in premature mortality due to rail emissions is partially counteracted by growth in diesel fuel use by the Indian rail sector, with 400 (95% CI: 300 to 600) additional annual premature mortalities in India. Furthermore, while measures such as electrification transfer combustion from the rail sector to the energy sector, in the case of countries such as China, total energy sector emissions still fall due to the effects of new emissions controls and other improvements applied to the local energy sector.

Reductions in emissions from the road transportation sector result in 4600 (95% CI: 3800 to 6000) fewer premature mortalities in 2015 compared to 2010 across the region. Changes in PM_{2.5}-driven premature mortalities from the road transportation sector are more

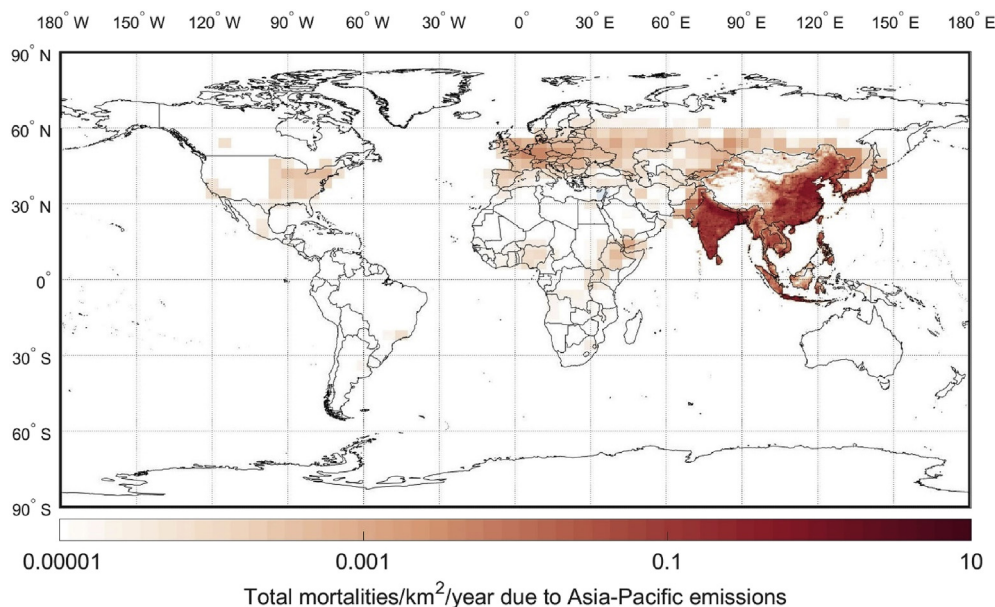


Fig. 7. Annual global premature mortalities due to total Asia-Pacific anthropogenic emissions in 2015 in units of mortalities/km²/year. Note logarithmic scales.

uniform across the region, as most countries, with the exception of Southeast Asian countries (as described in Fig. 2) other than Thailand, implemented more stringent vehicle emission and fuel quality standards for both gasoline and diesel fuels. This ranged from Euro II/III-equivalent standards in some countries (e.g. Indonesia and India respectively) in 2010, to Euro IV/V/VI-equivalent standards in most countries in 2015 (except for previously-noted countries in Southeast Asia, which maintain Euro II/III-equivalent standards). The improvements in emission and fuel quality standards not only offset all growth in emissions that might have occurred as a result of activity growth in this sector over the five-year period, but also result in up to 3000 (95% CI: 2600 to 4000) and 1000 (95% CI: 800 to 1400) avoided deaths in China and India respectively.

3.3.2. Global impacts

The spatial distribution of annual premature mortalities associated with total Asia-Pacific anthropogenic emissions in 2015 is presented in Fig. 7 on a per unit area basis. The results for the Asia-Pacific region are obtained from fine resolution GEOS-Chem simulations at $0.5^\circ \times 0.667^\circ$ and are overlaid on results for the rest of the world obtained from the coarse $4^\circ \times 5^\circ$ resolution simulations. Results obtained for the Asia-Pacific region from the coarse resolution simulations are found to agree to within 20% with results obtained from fine resolution simulations. This error margin lies within the 95% confidence interval of results in this study.

Premature mortality impacts due to Asia-Pacific emissions are exported to four of the six other global regions, with negligible impacts in Oceania and South America. Asia-Pacific anthropogenic emissions in 2015 cause 40,000 (95% CI: 35,000 to 44,000) exported premature mortalities due to exposure to PM_{2.5} (2% of total impacts). Of these, over 28,000 (95% CI: 26,000 to 32,000), or 73%, are estimated to occur in the Middle East, predominantly in highly populated countries bordering the Asia-Pacific region, such as Pakistan. Lower burdens of 5700 (95% CI: 4900 to 6600) and 3600 (95% CI: 3100 to 4200) are calculated for CIS and Europe respectively, with impacts in North America and Africa of less than 1000 mortalities each. The coarse resolution of the GEOS-Chem results used to compute the exported impacts makes them unsuitable for more detailed analysis. However, some further analysis on the sectoral contributors to these impacts and their temporal evolution is presented in the Appendix.

3.4. Limitations and future work

The mortality results from this study are affected by multiple limiting uncertainties. Firstly, as was mentioned in the discussion on validation against MODIS data, the emission modeling approach adopted for this study does not consider the effect of change in spatial variation of emissions within any country (e.g. due to demographic shifts or spatial variation in abatement measures). This results in likely underestimation of the collocation of population and emission of PM_{2.5} precursors in 2015, which may lead to underestimation of mortalities, particularly arising from species with predominantly local impacts such as BC, OC, and other primary particulates. We are also limited by the accuracy of our simulation in reproducing observed surface PM_{2.5} concentrations.

Furthermore, spatial variation in the effectiveness of abatement measures is particularly expected to affect the results for China, where recent studies (Koukoulis et al., 2016) have demonstrated greater effectiveness in adoption of abatement technologies by the energy sector in the south of the country compared to the north. Therefore, application of abatement measures uniformly across the country as was done in this study leads to underestimation of PM_{2.5} precursor emissions and formation in the north, and overestimation in the south in 2015. Given the higher population of the south of the country compared to the north (ORNL, 2016), this suggests that mortality reductions achieved in China over the 5 year period from 2010 to 2015 due to introduced abatement measures are also underestimated in this study.

Secondly, the results for total mortalities are likely to be underestimates of the total regional burden of air quality-driven premature mortality for two main reasons. Firstly, the health impacts computed for this study are computed using integrated exposure-responses from 2016, and do not include the latest epidemiological data from high-dosage cohort studies conducted in China, which yield higher mortality rates due to exposure of PM_{2.5} by up to 14% (Burnett et al., 2018). Secondly, this study is dedicated to estimation of PM_{2.5}-driven premature mortalities and does not take ozone-driven premature mortalities into account. Past literature (Turner et al. 2016) suggests that long-term exposure to ozone may contribute significantly to the total air quality-driven mortality burden, particularly in areas where abatement measures are introduced. Furthermore a recent study by Maji et al. (2019) estimates up to 75,000 annual premature mortalities, or ~7.6% of the total national air quality-driven premature mortalities,

attributable to exposure to ozone in China in 2016. However, an extended analysis of ozone's contribution to premature mortalities on a regional basis is beyond the scope of this study. These results are also sensitive to the baseline mortality rates in each country, as lower baseline mortality rates mean that the same level of air pollution exposure will result in fewer mortalities (although these factors are coupled).

This study has quantified damages from PM_{2.5}-driven air quality degradation on human health using premature mortalities as a common metric to facilitate comparison with several previous studies for this region over an extended timespan. However, as a more comprehensive quantification of the total damages to human health incurred by this region due to air pollution becomes a driving interest for future work, other metrics and endpoints could also be considered, including years of life lost (YLL); disability-adjusted life years (DALYs), which would encapsulate the effect of morbidity in addition to mortality; or economic costs.

4. Conclusions

Overall, we estimate that global PM_{2.5}-driven premature mortalities associated with anthropogenic emissions from the Asia-Pacific region increase from 1,780,000 annually in 2010 to 2,030,000 annually in 2015. Although 91% of this change is due to demographic factors such as population growth, 9% of the net change is driven by changes in emissions. This includes countervailing trends among emissions from different sectors and in different countries. In South and Southeast Asian countries, especially in India and Indonesia, emissions of primary PM_{2.5} and PM_{2.5} precursors increased by 40–60% due to activity growth. Emissions in several East Asian countries, particularly in China and Japan, decreased by 30–40% due to changes in activity patterns and regulations including China's 2013 Action Plan on the Prevention and Control of Air Pollution. In both 2010 and 2015, 98% of the premature mortality impacts are found to occur within the Asia-Pacific region while the remaining 2% are exported to the rest of the world. More than 80% of the regional impacts occur in China and India.

The agricultural, industrial, and residential sectors are found to cause the most region-wide premature mortalities in 2015, accounting for 30%, 22%, and 25% of the regional total respectively. The energy and industrial sectors are the leading contributors to gross increase in regional premature mortalities between 2010 and 2015, contributing a combined 64,000 additional premature mortalities in India alone, or

over 65% of the total estimate of 99,000 additional premature mortalities.

The relatively small changes in regional-mean air quality for Asia-Pacific between 2010 and 2015 obscure much higher magnitudes of air quality improvement and degradation in individual countries. This suggests the need for greater understanding of shifts in air quality burden within the region, especially of transboundary transport and the source-receptor relationships of air pollutants in this region. Future work in this area would also help inform policymaking through evaluation of the transboundary effects of policy choices made by individual countries on other regional stakeholders. Furthermore, by assessing the relative burden of domestic and transboundary air pollution-driven health impacts, each individual receptor country's degree of control over the primary sources of impacts within its borders could be determined. This would characterize the potential effectiveness of national-level mitigation strategies and inform the requirements for broader, region-wide mitigation strategies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atmosenv.2019.116916>.

Appendix

Table A1

Existing estimates of annual premature mortalities due to exposure to outdoor air pollution in China and India

Emission Year	2000	2010	2013	2015
Source	Silva et al. (2013)	Lelieveld et al. (2015)	Hu et al. (2017)	Cohen et al. (2017)
China	1,252,000 (incl. Other East Asia)	1,360,000	1,300,000	1,108,000
India	515,000	650,000	–	1,090,000

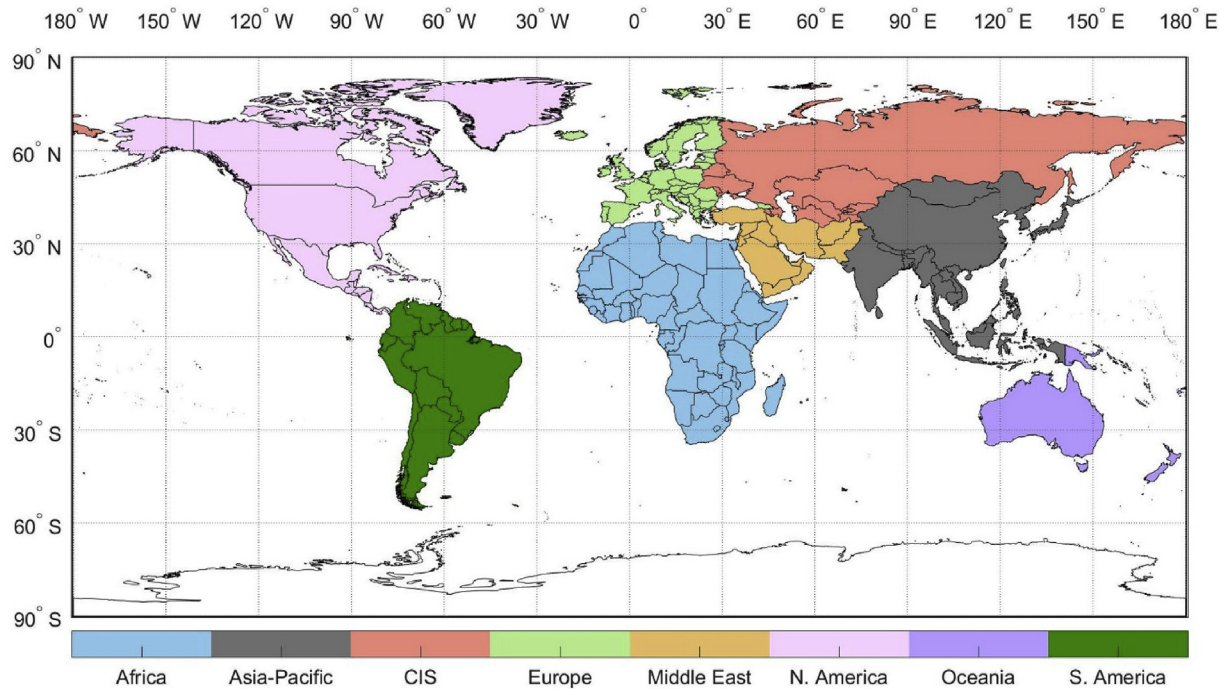


Fig. A2. Map of global regions as defined for this study.

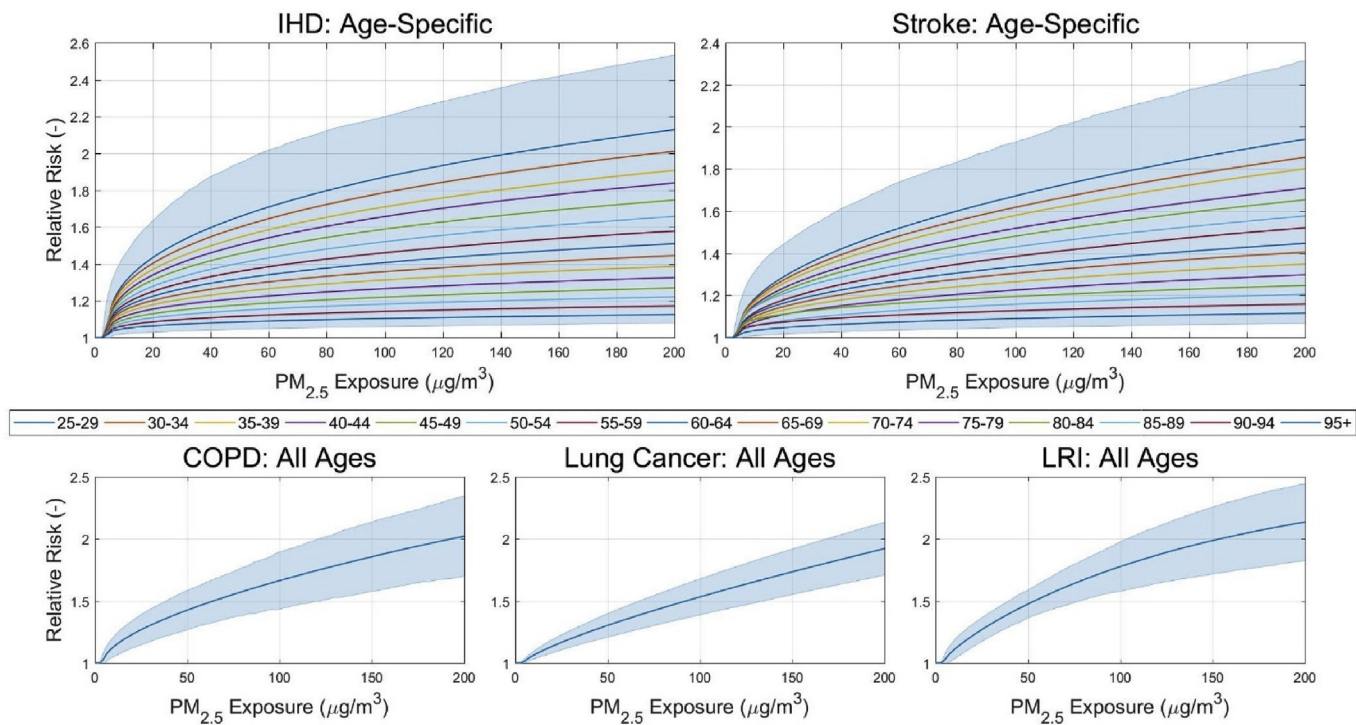


Fig. A3. WHO GBD 2016 IERs for COPD, IHD, LRI, lung cancer, and strokes (mean and 95% confidence intervals) (Gakidou et al., 2017).

Table A4

2010 and 2015 population-weighted ground-level PM_{2.5} in the Asia-Pacific region, as obtained from GEOS-Chem and NASA MODIS satellite observations in µg/m³

Receptor Region	Population-Wgtd. PM _{2.5} (2010)		Population-Wgtd. PM _{2.5} (2015)	
	MODIS	GEOS-Chem	MODIS	GEOS-Chem
Asia-Pacific	33.5	37.5	36.3	38.9
Bangladesh	42.7	38.0	53.5	45.3
China	39.2	50.9	38.7	44.8
India	41.8	37.4	47.5	45.4

(continued on next page)

Table A4 (continued)

Receptor Region	Population-Wgtd. PM _{2.5} (2010)		Population-Wgtd. PM _{2.5} (2015)	
Indonesia	7.0	18.3	11.7	26.7
Japan	7.8	11.4	9.2	10.7
Other	14.8	18.1	16.3	18.5

Table A5

Change in population-weighted PM_{2.5} in the Asia-Pacific region between 2010 and 2015, as obtained from GEOS-Chem and NASA MODIS satellite observations

Receptor Region	Change in Population-Wgtd. PM _{2.5} (MODIS)	Change in Population-Wgtd. PM _{2.5} (GEOS-Chem)
Asia-Pacific	2.8	1.4
Bangladesh	10.8	7.3
China	−0.5	−6.0
India	5.8	8.0
Indonesia	4.7	8.6
Japan	1.4	−0.7

Table A6

Sectoral contributions to total annual PM_{2.5}-driven mortalities in Asia-Pacific countries in 2015 in mortalities/year

Receptor Country	Agriculture	Energy	Fuel Processing	Industry	Residential	Rail Transport.	Road Transport.	Shipping
Bangladesh	21,000	12,000	100	8700	20,000	880	2800	900
China	250,000	100,000	49,000	230,000	220,000	4400	24,000	20,000
India	210,000	140,000	900	140,000	180,000	4200	25,000	2700
Indonesia	35,000	14,000	560	13,000	28,000	10	6600	770
Japan	14,000	5000	1100	12,000	4000	130	3700	2800
Myanmar	7800	1900	200	1800	4600	30	180	90
Philippines	8200	2500	860	3500	3800	0	890	730
South Korea	5000	2100	370	3200	2400	50	760	250
Thailand	8500	2600	2400	6400	4500	90	1500	260
Vietnam	10,000	3500	580	4900	11,000	40	1100	590
Others	15,000	9800	1400	12,000	14,000	180	2400	1100

Table A7

Sectoral contributions to change in annual PM_{2.5}-driven mortalities in Asia-Pacific countries between 2010 and 2015 in mortalities/year

Receptor Country	Agriculture	Energy	Fuel Processing	Industry	Residential	Rail Transport.	Road Transport.	Shipping
Bangladesh	1100	2000	10	2100	320	190	−90	80
China	6500	−30,000	3800	−25,000	−26,000	−1800	−3000	2900
India	6900	26,000	0	39,000	4200	430	−1000	0
Indonesia	3000	4500	−20	320	1900	10	910	−110
Japan	−250	−720	40	−1300	−410	−60	−390	270
Myanmar	60	220	20	240	−110	0	−40	10
Philippines	710	770	−50	290	170	0	10	70
South Korea	90	−190	20	−60	590	−20	−330	10
Thailand	650	30	50	−590	−140	−10	−190	30
Vietnam	960	420	60	−510	250	−20	−230	70
Others	990	−150	90	−680	−260	−20	−170	70

Sectoral breakdown and 2010–2015 change of exported impacts from the Asia-Pacific

Of sectoral contributors to global impacts, agriculture is dominant, resulting in 11,000 (95% CI: 9600 to 12,000) annual exported premature mortalities (Table A8). Energy, residential emissions, and industry all contribute between 6500 and 8800 exported mortalities in the baseline estimate. Combined with road transportation, these five sectors account for around 95% of the exported mortalities.

Table A8

Sectoral contributions to total exported annual PM_{2.5}-driven mortalities from the Asia-Pacific region in 2015 in mortalities/year

Receptor Region	Agriculture	Energy	Fuel Processing	Industry	Residential	Rail Transport.	Road Transport.	Shipping
Africa	100	200	20	130	160	10	10	20
CIS	2700	600	200	700	810	60	330	270
Europe	0	400	500	1000	700	100	700	200
Middle East	8100	7500	100	4500	7000	200	1500	100

(continued on next page)

Table A8 (continued)

Receptor Region	Agriculture	Energy	Fuel Processing	Industry	Residential	Rail Transport.	Road Transport.	Shipping
N. America	200	100	100	200	200	0	0	0
Oceania	0	0	0	0	0	0	0	0
S. America	0	0	0	0	0	0	0	0

The ranking of exported impacts also differs from the ranking of “in-region” impacts. Although industrial emissions have the greatest effect on mortality within the Asia-Pacific region, energy emissions are the greatest contributor to exported mortalities. This reversal can be attributed to two characteristics of the energy sector. Firstly, energy emissions include a higher proportion of secondary PM_{2.5} precursors, meaning greater impacts through long-range pathways of PM_{2.5} formation. Furthermore, energy sector emission sources are more prevalent in the Indian and Chinese border regions compared to industrial or residential sources.

The export of PM_{2.5}-driven premature mortalities from the Asia-Pacific region is also found to increase by 2500 (95% CI: 2300 to 3100) between 2010 and 2015 with 2800 additional premature mortalities exported to the Middle East and changes in all other regions of less than ± 300 mortalities each. The spatial distribution of the change in annual premature mortalities associated with total Asia-Pacific anthropogenic emissions between 2010 and 2015 is presented in Fig. A9 on a per unit area basis.

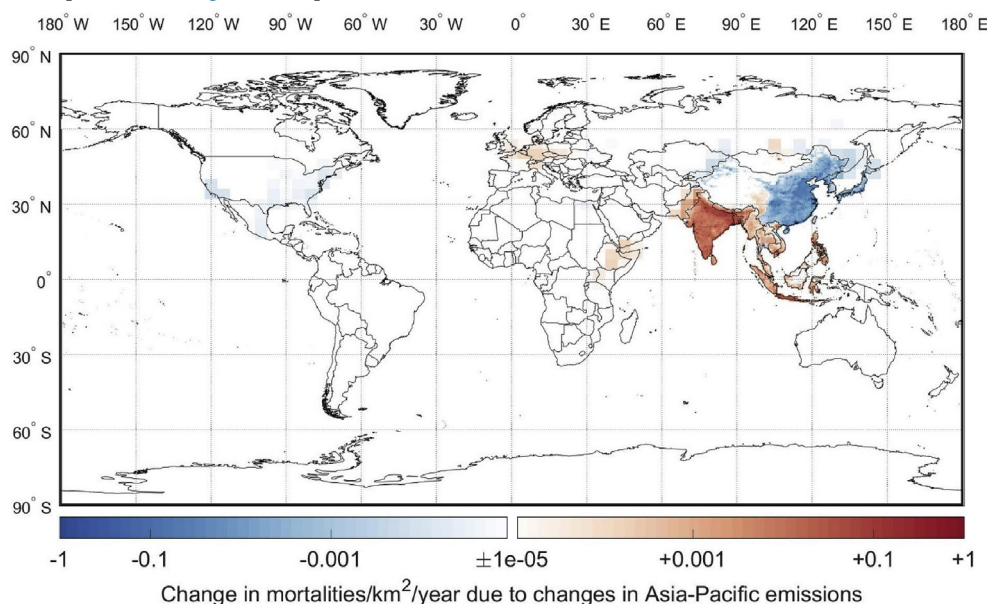


Fig. A9. Annual global premature mortalities due to change in Asia-Pacific anthropogenic emissions between 2010 and 2015, in units of mortalities/km²/year. Note logarithmic scales.

Sectoral contributions to growth in exported mortalities between 2010 and 2015 also differ from contributions to total exported mortalities in 2015. While the agricultural sector is the leading source of total exported mortalities in 2015, growth in Asia-Pacific energy and industrial emissions act as the primary driver of increases in exported mortalities, resulting in 1000 (95% CI: 800 to 1200) more mortalities exported in 2015 than in 2010 each. The growth of these two sectors in South Asia (mostly in India) is the primary driver of the increase in exported mortalities, as most of the additional impacts occur in the neighboring Middle East region. While the residential sector is the second largest contributor to total Asia-Pacific exported mortalities in 2015, it does not contribute to change between 2010 and 2015. This is also driven by the trend of emissions in South Asia, where residential emissions show little change between 2010 and 2015.

Emission controls applied across the Asia-Pacific region over this time period also affect exported mortalities. The effect of the controls applied to the energy, industrial, and residential sectors in China is discernible in changes to mortalities exported to the neighboring CIS region, where the energy sector causes 200 (33%) fewer annual exported premature mortalities in the CIS region in 2015 compared to 2010, and the industrial and residential sectors cause 100 (19%) and 13% respectively fewer premature mortalities each. In addition, improved vehicle emission and fuel quality standards imposed on Asia-Pacific road transportation result in 100 fewer exported mortalities from this sector in 2015 compared to 2010.

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